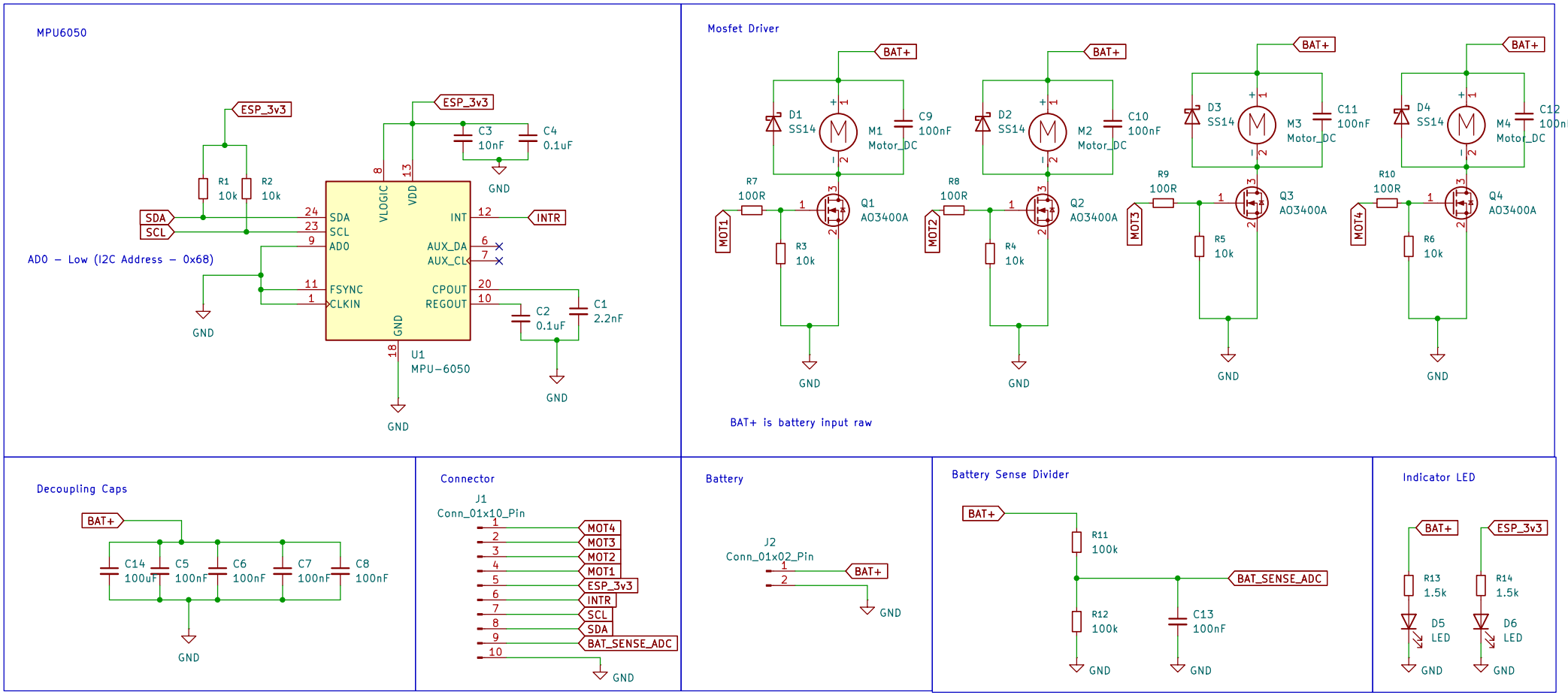
MPU6050
ESP_3v3
R1
10k
R2
10k
SDA
SCL
24
23
9
SDA
SCL
ADO
VLOGIC
8
VDD
13
ESP_3v3
C3
10nF
C4
0.1uF
GND
INT
12
INTR
AUX_DA
6
AUX_CL
7
ADO – Low (I2C Address – 0x68)
GND
11
1
FSYNC
CLKIN
CPOUT
20
REGOUT
10
C2
0.1uF
C1
2.2nF
GND
GND
18
U1
MPU-6050
Mosfet Driver
BAT+
D1
SS14
M1
Motor_DC
C9
100nF
R7
100R
MOT1
Q1
AO3400A
R3
10k
GND
D2
SS14
M2
Motor_DC
C10
100nF
R8
100R
MOT2
Q2
AO3400A
R4
10k
GND
D3
SS14
M3
Motor_DC
C11
100nF
R9
100R
MOT3
Q3
AO3400A
R5
10k
GND
D4
SS14
M4
Motor_DC
C12
100nF
R10
100R
MOT4
Q4
AO3400A
R6
10k
GND
BAT+ is battery input raw
Decoupling Caps
BAT+
C14
100uF
C5
100nF
C6
100nF
C7
100nF
C8
100nF
GND
Connector
J1
Conn_01x10_Pin
1
2
3
4
5
6
7
8
9
10
MOT4
MOT3
MOT2
MOT1
ESP_3v3
INTR
SCL
SDA
BAT_SENSE_ADC
GND
Battery
J2
Conn_01x02_Pin
1
2
BAT+
GND
Battery Sense Divider
BAT+
R11
100k
BAT_SENSE_ADC
R12
100k
C13
100nF
GND
GND
Indicator LED
BAT+
ESP_3v3
R13
1.5k
R14
1.5k
D5
LED
D6
LED
GND
GND

12 Oct 2026